DSAA 2012 Deep Learning

Project Guideline and Rubric

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1 Overview

The main goal of the course project is to produce and present a workshop-level work.

What Do We Expect? Each claim you make should be precise and supported. You can support a claim either by a citation, an experiment, or a mathematical argument. Be sparing with sentences that are subjective or arguable, and only include them with proper context. The course projects should go beyond re-implementing existing methods and instead seek new insights and understanding of topics discussed in class. It is not enough to do a literature review, nor to directly re-implement a method without making any changes. Projects can explore the theory or applications of deep learning. Application-oriented projects should ask questions that further our understanding of deep learning and not just your application area. For example, here are two possible applied projects, a mediocre one and a good one:

- ("Drug classification with deep nets") Apply deep nets to classifying molecules by their toxicity. Show that transformers get higher accuracy than RNNs on this problem. Achieve a high accuracy on the test set. [mediocre]
- ("Adaptive computation time transformers for molecular analysis") Identify that different molecules come in wildly different sizes, and should require different amounts of computation to analyze. Show how to modify a transformer to have depth or width that is adaptive to the size of the query molecule. Analyze how your system trades off time/memory for accuracy, and how properties of the molecule graphs affect this tradeoff. [good]

Projects related to your outside work You are welcome to work on a project that is related to your thesis topic or other outside interests. However, the work you do for this class should be novel to this class. You should not submit work for which you have gotten, or will get, credit toward another class or degree program. You should not seek assistance from any outside supervisors. You should be the single author for your project, with the assistance only from the instructor/GTAs.
Of course it's fine, and wonderful, if after this class you extend your project into a publication or blog post that you can share with the world. This is very likely, if you get a good grade for your project. It will be good to find some supervisor for your publication.

Formatting Use the NeurIPS preprint template with clear section headings, subheadings, and numbered sections for readability. Please download the NeurIPS LaTeX style file via the NeurIPS official website.

Page Limits The main text of a submitted paper is limited to four content pages, including figures and tables. Additional pages can only contain references and appendices. Note that, according to the NeurIPS/ICML/ICLR reviewing guideline, the reviewers are not required to read the appendices. DSAA 2012 follows this rule. The instructors and GTA may or may not read the appendices. So please make the main text strong and complete.

The Materials Which You Are Required to Submit

- Project Proposal (5 points): Submit a one-page pdf (the NeurIPS formatting).
  Get an early feedback for your project. Provide an outline of your plan for the project and questions you will investigate / analysis you'll conduct in the course of it.
- Project Report (35 points): Submit a paper (the NeurIPS formatting), with the maximum four-page main text and additional appendices.
  Produce a workshop-level paper that showcases a comprehensive understanding of the subject matter, significant innovation, meticulous attention to detail, and publishable quality, advancing the field.
- Video Presentation (20 points): Submit a recorded oral presentation video ($8 \sim 10$ min) and the presentation slides.
  Recording an oral presentation video for your project with the slides. It would be better to record your talking face with the video, suggested by top conferences oral presentation instructions.

Time Line These are hard dues. We have extended around two weeks for the project after the exam. In principle, no late submission will be allowed. Overdues will result in deducted or canceled scores.

1. Project Proposal: October 16th, 11:59 PM
2. Project Report: December 11th, 11:59 PM
3. Video Presentation: December 11th, 11:59 PM

2 Project Guideline

2.1 Project Proposal Guideline

Submit a proposal as a one-page pdf. Provide an outline of your plan for the project and questions you will investigate / analysis you'll conduct in the course of it. It may help to define a set of hypotheses you will test. An integral aspect of the proposal is to define a project idea that is both realistic and ambitious in scope. Its content can be considered as an extended version of the abstract with more details of introduction, research goals, research methods, research contributions, and expected research outcomes. We recommend that you use the project proposal stage to get feedback from the teaching staff on the project's feasibility and whether the proposal satisfies the project expectations of the class.

2.2 Project Report Guideline

Please refer to the following structured suggestions for the project report.

1. Abstract
   - Paragraph summarizing
     - Research problem, objectives, and key contributions.
     - Methodology (e.g., "We propose a diffusion model for anime avatar generation").
     - Main results (e.g., "Achieved FID score of 25.3 on the AnimeFace dataset").
     - Significance of findings.

2. Introduction
    - Context: Highlight the real-world or academic relevance of the problem.
        - Example: "Anime character design is critical for digital content creation but time-consuming."
    - Problem Statement: Clearly define the research question or hypothesis.
        - Example: "Can diffusion models generate high-quality anime avatars with user-specified attributes?"
    - Goals: List primary and secondary objectives.
        - Example: "Develop a model to generate diverse avatars with controlled features (e.g., hairstyle)."
    - Contributions: Concisely state novel aspects of your work.
3. Related Work
    - Literature Review: Summarize key studies relevant to your project.
        - Example: "Diffusion models have been widely used for anime generation, but struggles with fine-grained control."
    - Gap Analysis: Identify limitations in existing approaches.
        - Example: "Existing models lack conditional generation for specific attributes."
    - Positioning: Explain how your work addresses these gaps.
4. Methodology
    - The Proposed Method
        - Diagram and formal description of your method (e.g., diffusion model architecture).
        - Equations for core components (e.g., forward/backward diffusion processes).
    - Model Architecture
        - Introducing the used neural network architecture (e.g. Transformer).
    - Dataset and Preprocessing
        - Dataset source (e.g., AnimeFace Dataset) and preprocessing steps (e.g., resizing to 256x256).
        - Dataset collection (optional) will be an important contribution, if you make your own dataset.
    - Training Methods
        - Loss function (e.g., noise prediction loss), optimizer (e.g., AdamW), and hyperparameters.
5. Empirical Analysis
    - Experimental Setting
        - Experimental Design
        - Datasets
        - Evaluation Metrics: Quantitative metrics (e.g., FID, IS, precision-recall); Qualitative metrics (e.g., user surveys for attribute accuracy).
        - Implementation Details
    - Empirical Results
        - Quantitative Results
            * Tables/graphs comparing performance against baselines.
            * Example: "Our model achieves 20% higher attribute accuracy than the state-of-the-art."

  - Qualitative Results
    * Visual examples of generated avatars (e.g., figures with side-by-side comparisons).
    * User study outcomes (e.g., "85% of participants preferred our model's outputs").

6. Discussion

   - Interpretation of Results
     - Analyze trends (e.g., "Higher diversity in hair styles but lower accuracy in facial expressions").
     - Relate findings to the hypothesis.
   - Limitations
     - "Struggles with complex background generation."
   - Ethical Considerations
     - Bias in the dataset, privacy concerns, or potential misuse of generated content.

7. Conclusion

   - Summary: Restate key findings and contributions.
   - Future Work: Suggest extensions (e.g., "Integrate 3D pose estimation for dynamic avatars").
   - Broader Impact: Discuss real-world impact or applications.

8. Supplementary Materials/Appendix

   - Code: Critical parts of the implementation.
   - Extended Details/Results: Supplementary details or tables/figures not in the main report.
   - Graders may or may not read all supplementary materials, following the review guidelines for top conferences.

2.3 Presentation Guideline

Video Presentation Requirements

1. Format & Technical Specifications

   - MP4 format, 16:9 aspect ratio preferred, 1080p resolution (minimum 720p).
   - Audio must be clear, with no background noise.
   - Duration: 8–10 minutes (strict time limit; overages may result in score deductions).

2. Content Structure

   - Title & Abstract: Introduce the topic and research question.
   - Problem Statement: Clearly define the problem and its position/significance with key literatures.
   - Methodology/Approach: Explain technical details, algorithms, or experimental design.
   - Results & Analysis: Present findings with visualizations (charts, graphs, tables) and critical interpretation.
   - Conclusions & Future Work: Summarize contributions and outline next steps.

3. Delivery Expectations

   - Speaker should appear on camera (optional but recommended for engagement).
   - Maintain a professional tone; avoid jargon without explanation.
   - Use verbal cues (e.g., "Now let's look at the results") to guide viewers.

Slide Deck Requirements

1. Design & Formatting
   - Slides must be in PDF format, with consistent styling (fonts, colors, layout).
   - Limit text to bullet points (max 5–7 per slide); prioritize visuals (figures, diagrams) over walls of text.
   - Include citations for borrowed images or ideas.
2. Content Prioritization:
   - Highlight key messages on each slide (e.g., "Key Insight: X outperforms Y by 20%").
   - Technical details should be concise but accurate (e.g., equations should be readable and explained).
   - Avoid overloading slides; aim for 15 slides maximum (excluding references).

3 Grading Rubric

Five Metrics The project will be graded following the five major metrics separately: contribution, clarity, depth, literature review, and formatting. The overall grade will be calculated as the averaged value of the five metrics' scores.

Grading Order and Perspective We will 1) grade the proposal and provide early feedbacks, 2) grade the final report, and 3) grade the presentation, including videos and slides.

1. Project Proposal: We mainly evaluate the planned outcomes which a project is designed to achieve reflected in the proposal, following the Criteria below.
2. Project Report: We mainly evaluate the outcomes which a project has achieved and included in the report, following the Criteria below.
3. Video Presentation: We mainly evaluate the outcomes which a project has achieved and included in the video presentation materials, including a video and slides, following the Criteria below.

Criteria Grades are awarded based on the following structured criteria.

1. A+: Exceptional Distinction (95–100 points)
   - Overall: The report represents a high level of academic excellence. It is a top-workshop-level paper that showcases a comprehensive understanding of the subject matter, significant innovation, and meticulous attention to detail.
   - Contribution: Original work with publishable quality, advancing the field.
   - Clarity: Flawlessly structured, logically organized, and written at a top-workshop level.
   - Depth of Research and Knowledge: Extensive extra-curricular reading, critical analysis, and innovative interpretations.
   - Literature Review: Exhaustive engagement with relevant literature, critically synthesized to contextualize and justify contributions.
   - Formatting: Impeccable adherence to academic standards (e.g., citations, headings, figures)
2. A: Outstanding Distinction (90–94 points)
   - Overall: It has the potential to be published as top-workshop paper with minor revisions. It demonstrates a high level of competence, originality, and the ability to conduct in-depth research. However, there are still some minor areas that could be improved.

- Contribution: High-quality work with potential for publication.
- Clarity: Well-structured report with ignorable formatting/flow issues.
- Depth of Research and Knowledge: Significant extra-curricular engagement and critical thinking.
- Literature Review: Thorough review of key sources, but limited critical synthesis of niche areas.
- Formatting: Minor inconsistencies (e.g., citation style, figure labeling).

3. A-: Excellent Distinction (85–89 points)

   - Overall: It has the potential to be published as top-workshop paper with major revisions. It shows a good understanding of the research problem, a solid approach to solving it, and reasonable results, while there are some minor issues in clarity, correctness, or depth.
   - Contribution: Competent work addressing a clear gap.
   - Clarity: Clear structure but may lack detail in places (e.g., ambiguous diagrams).
   - Depth of Research and Knowledge: Good critical analysis and extra-curricular reading.
   - Literature Review: Adequate coverage of foundational works but limited engagement with recent or niche research.
   - Formatting: Occasional errors (e.g., inconsistent headings, misplaced references).

4. B: Merit (70–84 points)

   - Overall Quality: The work is of good quality, with a clear understanding of the basic concepts and an attempt to apply them. However, it lacks the depth, novelty, or completeness of the higher-graded work. There are areas that need improvement in terms of clarity, correctness, and depth, but it still shows potential for further development.
   - Contribution: Solid foundation but no novel insights.
   - Clarity: Logical structure but lacks coherence in sections (e.g., vague hypothesis).
   - Depth of Research and Knowledge: Demonstrates core knowledge with some critical thought.
   - Literature Review: Limited to core sources; minimal critical engagement or synthesis.
   - Formatting: Inconsistent style (e.g., mixed fonts, missing page numbers).

5. C: Borderline Pass (55–69 points)

   - Overall Quality: The work is of marginal quality. It shows some understanding of the core concepts but fails to demonstrate a comprehensive or in-depth approach. There are significant problems in clarity, correctness, and goal achievement, and it requires substantial improvement to reach a higher level of quality.
   - Contribution: Limited originality or including irrelevant content.
   - Clarity: Report is poorly structured, with unclear explanations.
   - Depth of Research and Knowledge: Basic grasp of core issues; limited extra-curricular engagement.
   - Literature Review: Superficial or outdated; fails to contextualize the work.
   - Formatting: Chaotic structure (e.g., missing sections, non-compliant citations).

6. D: Borderline Fail (40–54 points)

   - Overall Quality: The work is of poor quality. It fails to meet the basic requirements of a research project in terms of contribution, research depth, clarity, and correctness. While it may have some elements that could be salvaged, it requires additional work to meet pass standards.
   - Contribution: No meaningful contribution.

- Clarity: Report writing is sub-standard, with errors/omissions in key sections.
- Depth of Research and Knowledge: Clear gaps in knowledge; minimal extra-curricular reading.
- Literature Review: Minimal or absent; no connection to research context.
- Formatting: Largely non-compliant (e.g., no headings, uncited content).

7. F: Fail (0–39 points)

   - Overall Quality: The work is unacceptable. It shows a lack of understanding of the research process, a failure to meet the basic requirements of the project, and a significant absence of quality in all aspects. It requires a complete overhaul to reach a passing standard.
   - Contribution: No viable contribution. Incoherent, rushed, or missing critical elements.
   - Clarity: Incoherent, rushed, or missing critical sections (e.g., no methodology).
   - Depth of Research and Knowledge: Fundamental flaws in knowledge; no extra-curricular engagement.
   - Literature Review: Nonexistent.
   - Formatting: Entirely non-compliant or absent.

4 Project Ideas

This section provides a candidate list of project ideas. Lists of possible project examples are also available here and many past workshops. Note that these are guiding ideas and that you are encouraged to develop **your own ideas** outside of this list. Please try to design a project that truly motivates you, which will be a wonderful start to your academic career.
We provide project examples for the seven research directions.

1. Computer Vision

   - Garbage Classification System with Transfer Learning
     - Use ResNet or YOLO for object detection and develop a deployable model for smart waste sorting.
   - Simple Facial Expression Recognition System
     - Train a CNN on the FER-2013 dataset and integrate with OpenCV for real-time emotion analysis.
   - Low-Light Image Enhancement & Denoising
     - Apply generative models to improve image quality under low-light conditions.
   - Gesture-Controlled Drone Navigation
     - Generate control commands via MediaPipe hand keypoint detection and combine with traditional control algorithms for navigation.
   - Few-Shot Image Classification (Low-Resource Learning)
     - Train models on Mini-ImageNet using meta-learning (e.g., MAML) or few-shot techniques.

2. Natural Language Processing

   - Personalized Dialogue Chatbot
     - Fine-tune Hugging Face models to build a domain-specific chatbot supporting multi-turn conversations.
   - Text Sentiment Analysis & Visualization
     - Analyze social media text using BERT or RoBERTa and create interactive sentiment trend heatmaps.

- Poetry Generation Model
  - Train a Transformer-based model for Chinese/English poetry generation with customizable themes.
- Cross-Language Question Answering System
  - Integrate machine translation and QA models (e.g., T5) to enable multi-language question answering.
- Home Energy Consumption Forecasting
  - Model time-series data with LSTM/Transformer to predict 24-hour electricity usage.

3. Generative Models

- Anime Avatar Generator with Diffusion Model
  - Train a diffusion model, such as Stable Diffusion or a custom-built diffusion architecture, to generate personalized (anime) character avatars.
- Text-Driven Image Editing
  - Use Stable Diffusion to perform local image modifications (e.g., "change a cat to a dog").
- AI Music Generation
  - Generate genre-specific music clips using WaveNet or Diffusion models.

4. Medicine and Healthcare

- Pneumonia X-Ray Classification
  - Train a CNN on the Chest X-Ray Dataset for preliminary lung disease screening.
- Parkinson's Disease Speech Detection
  - Extract audio features and use LSTM models to identify Parkinson's-related speech patterns.
- Personalized Healthy Diet Recommender
  - Combine user input and Food101 image recognition for multi-modal nutrition suggestions.
- Medical Imaging Diagnosis Assistance System
  - Develop multi-modal fusion models for medical image analysis and integrate Grad-CAM for interpretable diagnosis.

5. Video and Audio

- Speech Emotion Recognition System
  - Train an LSTM+CNN hybrid model on the RAVDESS dataset to analyze emotional tone.
- Short Video Captioning and Tagging
  - Automatically label short videos with descriptions using CLIP models.
- Video Super-Resolution
  - Enhance low-resolution videos using EDSR or Diffusion Models.
- Deep Learning-Based Music Recommendation
  - Build a personalized recommender using user behavior data and audio features (e.g., MFCC).

6. Deep Learning Understanding and Analysis

- Research on Visualization of Convolutional Neural Network (CNN) Feature Maps

  - Analyze the feature extraction process of CNNs on CIFAR-10/MNIST through Grad-CAM and feature map visualization tools, and explore the semantic levels of different layers.
- Analysis of the Attention Mechanism in Transformer Models
  - Visualize the attention weights of Transformer models in text classification/translation tasks using the GLUE dataset, and compare the attention patterns of different heads.
- Research and Optimization of Neural Network Pruning Mechanisms
  - Implement structured/unstructured pruning on LeNet/AlexNet, and analyze the impact of the pruning rate on the accuracy and inference speed of MNIST/CIFAR-10 models.
- Analysis of the Latent Space of Variational Autoencoders (VAEs)
  - Train VAEs to generate MNIST/CIFAR-10 images. Visualize the structure of the latent space through t-SNE/UMAP, and study interpolation experiments and anomaly detection capabilities.
- Research on Visualization of the Denoising Process in Diffusion Models
  - Train a diffusion model on the CelebA dataset. Visualize the reverse process of gradually eliminating noise and analyze the generation patterns at different time steps.
- Comparative Study of Interpretability Methods for Deep Learning Models
  - Use LIME, SHAP, and DeepLIFT to explain the predictions of image classification (ResNet) and text classification (BERT) models, and compare the accuracy and efficiency of these methods.

7. AI for Science

- Protein Structure Prediction and Function Analysis
  - Use AlphaFold2 to predict 3D protein structures and analyze biological functions with sequence features.
- Chemical Synthesis Path Optimization System
  - Train sequence models to generate synthesis steps and integrate reinforcement learning to optimize yield and safety.
- Climate Change Ecosystem Simulation
  - Build species distribution models and multi-agent systems to simulate the impact of climate change on biodiversity.
- Astronomical Spectrum Classification System
  - Utilize diffusion models to generate spectral data and combine transfer learning for automatic stellar